

1. Administrative & Study Identification

Full Study Title: Modular Clinical Pharmacology Study to Evaluate the Drug-drug Interaction Potential and Relative Bioavailability of Saruparib **Short Title/Acronym:** Saruparib DDI and Relative Bioavailability Study **Protocol Number:** [Internal Study ID] **NCT Number:** NCT06899061 **Posting ID:** 220301648810634864 **Sponsor Name:** AstraZeneca **Investigational Product:** Saruparib (oral) **Phase of Development:** Phase 1 **Trial Status:** Recruiting **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the drug-drug interaction (DDI) potential of oral saruparib on the pharmacokinetics (PK) of human drug transporter substrates (digoxin, furosemide, metformin hydrochloride, and rosuvastatin) in participants with advanced solid malignancies (Module 1), and to evaluate the relative bioavailability and proton pump inhibitor (PPI) effect of saruparib tablets manufactured using a direct compression (DC) process (Module 2). **Secondary Objectives:** To evaluate the overall safety, tolerability, and clinical activity of saruparib in the target oncology population.

2.2 Background & Rationale To address the pharmacological profile of saruparib in patients with advanced solid malignancies. As patients routinely require concomitant medications, this study generates essential data on how saruparib impacts major drug transporters (P-gp, OAT1/3, OCT2/MATE1/2K, and OATP1B1/3) and how its bioavailability is affected by formulations and gastric pH modifiers (like rabeprazole) to ensure safe co-administration.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Clinical Pharmacology) **Control Type:** Single-arm (Module 1); Active Comparator/Crossover (Module 2) **Blinding:** Open-label **Randomization:** N/A (Module 1); Randomized crossover (Module 2)

3.2 Planned Enrollment & Duration Target \$N\$: 45 Number of Sites: Multi-centre **Estimated Study Start:** March 2025 **Estimated Primary Completion:** [DD-MMM-YYYY]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants with documented evidence of locally advanced unresectable or metastatic solid tumours, excluding lymphoma, who have exhausted standard of care options (or for which no standard therapies exist) and may be suitable for saruparib monotherapy treatment in the opinion of the investigator. 2. Adequate organ and marrow function (in the absence of transfusions or growth factor support within 28 days prior to enrolment). 3. Eastern Cooperative Oncology Group (ECOG) Performance Status 0 to 1 with no deterioration within 1 week prior to the screening visit. 4. Life expectancy ≥ 12 weeks.

4.2 Exclusion Criteria 1. Evidence of active and uncontrolled hepatitis B and/or hepatitis C. 2. Participants with controlled HIV who do not meet specific viral load and CD4+ count stability criteria. 3. Any other evidence of severe or uncontrolled systemic diseases or active uncontrolled infections, including major seizure disorder, active bleeding diseases, superior vena cava syndrome, or history of allogenic organ transplant. 4. Active tuberculosis infection. 5. Cardiac criteria including mean resting QTcF > 470 ms, factors increasing the risk of QT prolongation/arrhythmia, or history of symptomatic/requiring treatment arrhythmia. 6. Other cardiovascular diseases (except alopecia and peripheral neuropathy) or unresolved toxicities $>$ CTCAE Grade 1 from prior anticancer therapy. 7. Any known history of persisting (> 2 weeks) severe pancytopenia ($ANC < 0.5 \times 10^9/L$ or platelets $< 50 \times 10^9/L$). 8. Spinal cord compression, or brain metastases for at least 4 weeks prior to study intervention unless asymptomatic and stable. 9. Known predisposition to bleeding (e.g., active peptic ulceration, recent haemorrhagic stroke, proliferative diabetic retinopathy). 10. History of Myelodysplastic Syndromes (MDS) or Acute Myeloid Leukaemia (AML), or with features suggestive of MDS/AML. 11. Refractory nausea and vomiting, chronic GI diseases, inability to swallow the formulated product, or previous significant bowel resection. 12. Known allergy or hypersensitivity to study intervention or excipients. 13. Past medical history of ILD, drug-induced ILD, radiation pneumonitis requiring steroids, or clinically active ILD. 14. Uncontrolled intercurrent illness within the last 12 months, including serious chronic GI conditions or psychiatric illness. 15. Current or planned use of calcium channel blockers up to the end of Period 2 (Module 1). 16. Current and/or regular use of any of the interventions in the cocktail substrates that cannot be substituted (Module 1).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Module 1:* Period 1: Single oral dose of a substrate cocktail (digoxin, furosemide, metformin, rosuvastatin) on Day 1. Period 2: Continuous saruparib from Day 1 to 9, with the cocktail co-administered on Day 5.

Period 3: Daily saruparib for up to three 28-day cycles. * *Module 2*: Single doses of roller compaction (RC) or direct compression (DC) saruparib tablets. Period 3 evaluates co-administration with rabeprazole. **Route of Administration:** Oral **Duration of Treatment:** Up to 3 cycles of 28 days each (for standard Saruparib monotherapy continuation).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care (excluding specific inhibitors/inducers). **Prohibited:** Current and/or regular use of any of the interventions in the cocktail substrates that cannot be substituted; calcium channel blockers during PK periods.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Medical History, Baseline Oncology Assessments
2	1 to 9	Cocktail substrate dosing (Module 1) / RC or DC saruparib dosing (Module 2); PK Labs
3	1 to 3	Washout between Period 1 and Period 2
4	1 to 9	Saruparib continuous dosing + Day 5 combination PK evaluation
5	Up to 3 Cycles	Monotherapy continuation; Safety & Tolerability assessments
6	+3 Days	Final Safety follow-up up to 3 days after the last dose

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: * Area under plasma concentration-time curve from zero extrapolated to infinity (AUCinf) of digoxin, furosemide, metformin and rosuvastatin when dosed alone and in combination with saruparib. * Area under the plasma concentration curve from zero to the last quantifiable concentration (AUClast) of digoxin, furosemide, metformin and rosuvastatin. * Maximum observed plasma (peak) drug concentration (Cmax) of digoxin, furosemide, metformin and rosuvastatin. * Relative bioavailability PK parameters of DC versus RC saruparib (Module 2). **Measurement Method:** Central Laboratory Analysis of plasma concentrations.

7.2 Secondary & Exploratory Endpoints * Time to reach peak or maximum observed concentration (tmax) of digoxin, furosemide, metformin and rosuvastatin. * Half-life associated with terminal slope (t1/2λz) of digoxin, furosemide, metformin and rosuvastatin. * Apparent total body clearance of drug from plasma after extravascular administration (CL/F) of digoxin, furosemide, metformin and rosuvastatin. * Incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs). * Objective Response Rate (ORR) or preliminary anti-tumour activity parameters in participants continuing saruparib treatment.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous Phase I clinical pharmacology monitoring and laboratory parameter tracking. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal AstraZeneca safety monitoring team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set for primary DDI/bioavailability endpoints; Safety Set for all enrolled patients receiving at least one dose. **Sample Size Justification:** A target enrollment of 45 participants is sized to provide adequate power for observing intra-subject differences in PK area under the curve (AUC) and Cmax in a crossover/sequential Phase I design. **Handling of Missing Data:** PK parameters will not be imputed; available data will be summarized using standard non-compartmental analysis (NCA).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring for Phase 1 PK compliance. **Audit Trail:** Validated eCRF systems with real-time data clarification tracking.

11. Study Identifiers & Governance

Primary ID: [AstraZeneca Internal ID] **Registry Number:** NCT06899061 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Modular Clinical Pharmacology Study to Evaluate the Drug-drug Interaction Potential and Relative Bioavailability of Saruparib **Official Regulatory Title:** A Modular Phase I, Open-label Study to Assess the Safety, Pharmacokinetics, and Drug Interaction Potential and Relative Bioavailability of Saruparib in Patients With Advanced Solid Malignancies

12. Clinical Framework (Oncology Specific) *

Primary Condition: Advanced Solid Malignancies **Diagnosis Threshold:** Documented evidence of locally advanced unresectable or metastatic solid tumours **Medical Methodology:** Patients must have exhausted standard of care options **Optimization Target:** Saruparib PK characterization and safety confirmation

13. Study Procedures & Methodology *

Management Pathway: Phase I Pharmacokinetic / DDI / Bioavailability Assessment **Education Protocol (HCP):** Site initiation visit & PK protocol training **Education Protocol (Patient):** Dosing compliance & specialized PK fasting/feeding instructions **Quality Audit Mechanism:** Rigorous tracking of PK sample collection windows and handling

14. Observation & Follow-Up Schedule *

Total Duration: Up to 12 weeks minimum (including 3 cycles of 28 days) **Onsite Visit Cadence:** Intensive daily sampling during Period 1 and Period 2; regular cyclic visits during Period 3/4 **Remote Monitoring Frequency:** Between cycles and End of Study follow-up **Checklist Review Interval:** Daily during PK admission periods

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					

(Note: Sections 12-14 have been appropriately adapted from the source template's Hyperkalemia defaults to reflect the relevant parameters of this Advanced Solid Malignancies oncology trial).